

The Welfare Gate: Pre-Fisc Inequality, Universal Public Services, and Democratic Resilience in Post-1990 Democratizers

A pre-registered research design. No model has been estimated; the estimates do not exist yet. The protocol, the fixed country roster, the coverage matrix, and the power analysis are deposited in a timestamped registry (OSF; the registration ID and deposit date are fixed at the moment of deposit) before any estimation. This paper states the theory, the data infrastructure, the six pre-committed estimation designs, and the decision rules against which results will be judged. It is written for venues that publish registered designs and pre-analysis plans; we do not report estimates because none are in hand.

Abstract

The literature on democratic erosion in the twenty-first century holds two contradictory accounts. The materialist account finds that income inequality is among the strongest predictors of democratic decline in a global sample of uninterrupted democracies [Rau and Stokes, 2025]. The coalitional account argues, from close case comparison, that economic grievances did not systematically separate post-1990 backsliders from survivors [Haggard and Kaufman, 2021]. This paper argues that the contradiction is unresolved for a structural reason: neither account tests the conditioning structure, and the statistical account’s treatment is entangled with the very institution the gate hypothesis implicates. Disposable-income inequality is post-fisc, and fiscal systems that compress it are larger where welfare states are larger, so a gate estimated on the disposable Gini cannot separate institutional transmission from arithmetic filtering. We register a design that corrects both problems: the treatment is pre-fisc, the World Inequality Database pre-tax top-10 percent income share, with the market-income Gini secondary. The fiscal filter is measured directly as the market-minus-disposable Gini gap, converting the entanglement into the test of the stabilization mechanism. Among post-1990 democratizers whose transition began between 1989 and 1998, the hypothesis is that market-inequality surges erode the V-Dem Liberal Democracy Index three to five years later only where universal public health and education services built in the first democratic decade are thin. The gate is identified off deviations from welfare-specific trends, with estimators built for differential treatment timing, a fixed roster with a documented coverage matrix, and a simulation-based minimum-detectable-effect analysis. Confirmation and disconfirmation criteria are fixed ex ante, with a primary-specification significance rule and a five-of-seven robustness threshold. The design is built so the test can adjudicate between unconditional materialist, purely coalitional, and institutionally gated accounts.

1 Introduction

Since the late twentieth century the main threat to democratic regimes has changed character. The military coup has given way to the power-aggrandizing elected executive: presidents and prime ministers who capture courts, harass the press, and de-legitimize elections [Rau and Stokes, 2025, Waldner and Lust, 2018, Bermeo, 2016, Levitsky and Ziblatt, 2018]. Erosion is now understood as a

graded, incremental process inside formally democratic regimes, and a third wave of autocratization is dated to the late 2000s [Lührmann and Lindberg, 2019]. The process descends from national executives to state governments [Grumbach, 2023]. The question that dominated twentieth-century comparative politics, what kills democracies, has been replaced by a harder one: how much democratic quality does a regime lose, and which democracies lose almost none?

Two answers compete. The materialist account points to income inequality. In a cross-national sample of democracies that remained uninterrupted over 1995–2020, Rau and Stokes count 23 spells of backsliding in 22 countries, lasting nine years on average, and find that the Gini coefficient is among the strongest predictors of erosion onset [Rau and Stokes, 2025]. The result extends an older one: in the coup era, inequality harmed democratic consolidation even where it did not block democratization [Houle, 2009], in line with the distributional-conflict tradition in which redistribution is the pivot of regime conflict [Acemoglu and Robinson, 2006, Boix, 2003, Ansell and Samuels, 2014]. The coalitional account pushes back. From close comparison of the post-1990 wave, Haggard and Kaufman conclude that economic grievances and rising inequality did not systematically separate backsliders from survivors; polarization and elite coalitional choices did [Haggard and Kaufman, 2021, Svoboda, 2019]. One literature says inequality is a central driver of erosion. Another says the economic story fails exactly in the cases that matter.

Both claims cannot hold in the unconditional form in which they are stated. The dispute has not been resolved because neither side runs the design that could resolve it. The statistical account estimates a main effect: its specifications contain no welfare variables and no moderators, and its sample of uninterrupted democracies excludes precisely the young democratizers that fell below the democracy threshold while eroding [Rau and Stokes, 2025]. The case-based account compares backsliders with survivors but never formalizes the interaction structure a conditional claim requires [Haggard and Kaufman, 2021]. What is missing in both is the institutional environment, and in the statistical account there is a deeper problem. Rau and Stokes measure inequality with disposable-income Ginis and with World Inequality Database shares that are pre-tax [Rau and Stokes, 2025]. Disposable income is post-fisc. In any country the fiscal system compresses market inequality, and the compression is larger where public services and transfers are larger. A specification that uses the disposable Gini as the treatment, with no welfare variables, cannot separate “inequality erodes democracy” from “fiscal systems that compress inequality also protect democracy.” The first mechanism of any welfare-gate account, automatic stabilization, is built into the treatment of the very paper that establishes the unconditional result.

This paper registers a design that corrects both problems and states the gate hypothesis plainly. Among post-1990 democratizers whose transition to electoral democracy began between 1989 and 1998, a rise in market-income inequality, measured pre-fisc, erodes liberal-democratic quality three to five years later only where welfare institutions built in the first democratic decade are thin. In countries whose first democratic decade universalized public health and education into services the broad middle class actually uses, market-inequality surges do not transmit into declines in the Liberal Democracy Index; where public services stayed thin, the same surges predict graded erosion. The gate decomposes into two paths, and the design measures them separately. Automatic stabilization is the fiscal filter: the market-minus-disposable Gini gap, the share of any market shock that the fiscal system absorbs before it reaches the distribution citizens experience. Behavioral transmission is the residual link from the market shock citizens do experience to democratic quality, gated by constituency formation, by which the middle class holds a material stake in the public institutions erosion would capture [Pierson, 1993, Soss and Schram, 2007, Rothstein and Usaner, 2005], and by credible pre-commitment, by which universal automatic programs block the elite-retrenchment and populist-promise equilibrium of Acemoglu, Egorov, and Sonin [Acemoglu et al., 2013]. The gate is the universal service state, not welfare effort in general: pension-heavy total social spending,

targeted cash, and chauvinist transfers do not gate, and generosity purchased later by incumbents is part of the erosion playbook, not a substitute [Szikra and Öktem, 2023, Lendvai-Bainton and Szelewa, 2021, Vanhuysse, 2006, Benczes and Orzechowska-Waclawska, 2024, Ennser-Jedenastik, 2018].

Why has this design not been run before? Three literatures hold its parts, and none has assembled them. The welfare-state literature established the concepts the gate needs: decommodification, by which welfare institutions sever living standards from labor-market position [Esping-Andersen, 1990]; the paradox of redistribution, by which universal programs redistribute more than targeted ones [Korpi and Palme, 1998]; and policy feedback, by which programs create their own defenders [Pierson, 1993, Soss and Schram, 2007]. The democratic-dividend program established that democratization raises tax revenues and schooling, and found no robust average effect on inequality, with imprecise estimates on net and gross Gini coefficients and suggestive heterogeneity by land inequality and structural transformation [Acemoglu et al., 2015]. The erosion literature measures neither welfare institutions nor the post-1990 cohort’s conditional structure [Rau and Stokes, 2025, Haggard and Kaufman, 2021]. The closest existing statements of the gate’s logic are the finding that young democracies are distinctively vulnerable to incumbent takeover rather than coups [Svolik, 2015], the argument that welfare states underwrite democratic legitimacy and consolidation in Latin America [Huber and Stephens, 2001, 2012], and the universalism-to-trust microfoundation of institutional legitimacy [Rothstein and Uslaner, 2005]. Each stops short of the inequality-shock interaction structure this design estimates.

The status is stated plainly because it matters for how the paper is read. No model has been estimated. This is a pre-registered research design: the roster, the coverage matrix, the six estimation designs, and the decision rules in Section 6 are fixed and deposited before estimation. We present the theory, the data infrastructure, and the protocol; a companion paper will report estimates against the Section 6 criteria. The design is restricted to public Our World in Data (OWID) panels [Our World in Data, 2025], with one disclosed non-OWID input, the Boix-Miller-Rosato cross-check for transition dating [Boix et al., 2013]; every variable is downloadable by any reader, and the full analysis is reproducible without proprietary data.

Summary of contributions. (i) We reconcile the materialist and coalitional accounts of post-1990 erosion by identifying the institutional condition under which each holds: inequality erodes democracy where welfare institutions are thin, and the coalitional story dominates where they are thick. (ii) We correct the measurement entanglement that has biased the materialist account toward a spurious unconditional effect: the treatment is pre-fisc, and the fiscal filter is measured directly as the market-minus-disposable Gini gap, so the stabilization mechanism is tested rather than assumed. (iii) We condition the inequality leg of the democratic dividend on welfare institutions [Acemoglu et al., 2015], state the dividend on top-income shares over a fixed post-transition window, and link its size to regime resilience two decades later. (iv) We identify the gate off deviations from welfare-specific trends, with estimators designed for differential treatment timing [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021], exogeneity probes on the pre-determined welfare measure, and a fixed roster with a documented coverage matrix, so the confirmation criteria are calibrated to the power the sample can actually deliver.

2 The Two Accounts and the Missing Interaction

2.1 The materialist account

Rau and Stokes identify erosion spells using expert-survey measures of decline in vertical and horizontal accountability, coded from V-Dem information [Rau and Stokes, 2025]. Between 1995 and

2020 they count 23 spells of backsliding in 22 countries, lasting nine years on average. The unit of analysis is the country-year, and inclusion requires a country to have been an uninterrupted democracy over the period under the Miller-Boix-Rosato classification; the main analyses cover 92 countries, with Venezuela the single admitted case that eroded into autocracy [Rau and Stokes, 2025, Boix et al., 2013]. The leading predictor is the Gini coefficient, and the calibration is stark. For Sweden, whose Gini of 26.4 in 2017 makes it more equal than 87 percent of democracies, the predicted annual risk of erosion is 4 percent. For the United States, whose Gini of 38.4 exceeds that of 60 percent of democracies, the predicted risk is 8.4 percent. South Africa, the most unequal democracy in the sample, has a predicted risk of 31 percent [Rau and Stokes, 2025]. The result extends to the structure of inequality: the larger the income and wealth shares of the top 1 and top 10 percent, the more likely democracy is to erode, and the larger the bottom-half share, the less likely [Rau and Stokes, 2025]. The association survives a wide battery of alternatives: four Gini sources, wealth inequality, country-year and country-election-year units, rare-events estimation, and region and country fixed effects [Rau and Stokes, 2025].

Two details of that paper matter for this one. First, the top-end income and wealth shares come from the World Inequality Database, and those shares are pre-tax [Rau and Stokes, 2025]. The flagship result’s structural measures are already market-income in concept, while its Gini measures are post-fisc, and the design estimates a single main effect pooling both. Second, Rau and Stokes interpret the channel through polarization: their mechanism analysis uses a measure of affective polarization, with V-Dem’s societal polarization as the robustness alternative [Rau and Stokes, 2025, Svulik, 2019]. The gate account does not contest that polarization is the proximate channel; it contests the claim that polarization is a sufficient channel, because the polarization-to-erosion link is exactly what universal service institutions are hypothesized to interrupt. We return to this in Section 3 and pre-commit the differentiating reading in Section 6.

The account’s lineage matters for what it does not test. Houle’s result, that inequality harms democratic consolidation but not democratization, was estimated in a global sample across the coup-driven era with discrete survival outcomes [Houle, 2009]. Haggard and Kaufman’s earlier statistical work likewise found inequality among the correlates of democratic instability [Haggard and Kaufman, 2012]. The theoretical tradition behind both is the distributional-conflict model of regimes, in which democratization is a commitment device that elites accept when the revolution constraint binds, and the redistributive threat it poses is what makes elites resist [Acemoglu and Robinson, 2006, Boix, 2003, Ansell and Samuels, 2014]. In that tradition inequality is the central cause, and welfare policy is a consequence of democracy, not a condition on its survival.

2.2 The coalitional account

Haggard and Kaufman’s comparative work on the post-1990 wave reaches a different conclusion [Haggard and Kaufman, 2021, 2016]. Backsliders such as Hungary, Poland, Turkey, and Venezuela, and survivors such as the Baltic states, Slovakia, South Korea, and Chile, do not separate cleanly on economic grievances. Some backsliders grew rapidly and reduced poverty; some survivors suffered deep crises. What distinguishes the two groups, in their reading, is the configuration of elites and parties: whether the political system produced a dominant coalition with incentives to dismantle constraints, and whether polarization gave that coalition an electorally sustainable base [Haggard and Kaufman, 2021, Svulik, 2019]. Their empirical strategy is comparative case analysis, and they are explicit that the economic variables that dominate the statistical literature do not carry the explanatory weight in the cases [Haggard and Kaufman, 2021]. Bermeo’s conceptual history and Waldner and Lust’s synthesis stress steady, incremental, often stealthy executive aggrandizement as a political choice rather than an economic reflex [Waldner and Lust, 2018, Bermeo, 2016].

Lührmann’s account of autocratization as a sequence locates the disruptive potential in institutional resilience, the general phenomenon of which the welfare state is, in our account, a specific instance [Lührmann, 2021].

2.3 Prior conditioning results

Three bodies of work already touch the conditional structure the gate needs, and each stops short of it. We state the differences now because the gate’s contribution is defined against them.

Young democracies are the population at risk. Svobik shows that young democracies are vulnerable to incumbent takeovers rather than coups, and that consolidation is slow: most democratic failures in the postwar era are executive power grabs in countries that have not yet experienced successful alternations in power [Svobik, 2015]. This defines the population this paper studies, the post-1990 cohort, and the outcome class, executive-aggrandizing erosion. Svobik conditions the risk on the history of electoral contestation, not on welfare institutions, and his framework does not engage the inequality channel at all. The gate adds exactly that: within the young-democracy population, the inequality-erosion link is conditional on the service state built in the first democratic decade.

Welfare states underwrite democratic legitimacy and consolidation. Huber and Stephens argue, from Latin America and from the advanced democracies, that welfare states provide legitimacy to democratic institutions and that their development and crisis track the fate of democracy [Huber and Stephens, 2001, 2012]. This is the closest existing statement of the gate’s constituency logic. It differs from the gate in three ways: it is about average legitimacy rather than the effect of inequality shocks; it does not test an interaction between welfare structure and inequality movements on democratic trajectories; and its consolidation accounts concern longer-lived democracies, not the post-1990 cohort with its inherited service legacies.

Universalism produces the legitimacy that protects institutions. Rothstein and Uslaner establish the microfoundation: universal welfare institutions generate equality, equality generates generalized trust, and trust generates support for institutions [Rothstein and Uslaner, 2005]. The gate’s constituency-formation mechanism is this argument applied at the margin of an inequality shock: universal services give the middle class a material stake in the public institutions that erosion would capture, so the institutions themselves acquire defenders [Pierson, 1993, Soss and Schram, 2007, Rothstein and Uslaner, 2005]. The gate claims effect modification, a conditional claim none of these works tests.

2.4 The missing interaction and the dividend’s null leg

Neither leading account tests the conditioning structure. The statistical account has no interaction terms: its specifications include no welfare variables, so it cannot observe whether the inequality coefficient is carried by the thin-welfare democracies in its sample, and its sample restriction excludes exactly the case population that matters most for adjudication, the young democratizers that fell below the democracy threshold while eroding [Rau and Stokes, 2025]. The case-based account has an interaction design in substance, backsliders versus survivors across institutional environments, but no welfare measure is coded systematically and no statistical claim about effect modification is made [Haggard and Kaufman, 2021]. The two accounts are not in fact rivals at the level of their designs. They are main-effect and case-descriptive enterprises, each silent on the dimension that would separate them.

A third literature makes the conditional structure concrete, and its own null marks the place where the gate should sit. The democratic-dividend program established that democratization has

a robust effect on tax revenues, about a 16 percent long-run increase in the tax-to-GDP ratio in the preferred specification, and on secondary schooling and structural transformation, but no robust average effect on inequality [Acemoglu et al., 2015]. The chapter’s inequality analysis runs net and gross Gini coefficients and top- and bottom-income shares relative to the middle; some specifications show negative effects on inequality, especially on the gross Gini, but the estimates carry large standard errors and are not stable, and the heterogeneity it reports, by land inequality, structural transformation, and the top-bottom share structure, is suggestive rather than robust [Acemoglu et al., 2015]. We cite the chapter for what it reports: imprecise average estimates, not a robust top-share decline. The field fact to explain is therefore not why inequality fails to fall on average, but under what conditions it does fall and whether that condition also gates erosion. The gate hypothesis supplies one candidate condition, welfare institutions, and the design in Section 5 makes the dividend conditional in exactly the way the chapter’s own conclusion invites: it calls for a systematic investigation of the conditions under which democracy reduces inequality [Acemoglu et al., 2015].

The outcome-side literature sharpens the direction of the test. Szikra and Öktem document how welfare states are transformed under democratic backsliding in Hungary and Turkey, converting universal or semi-universal systems into instruments of the illiberal project [Szikra and Öktem, 2023]. Lendvai-Bainton and Szelewa trace the same conversion in Hungary and Poland’s radical welfare reforms [Lendvai-Bainton and Szelewa, 2021]. Vanhuyse reads early post-communist social policy as strategic demobilization of protest [Vanhuyse, 2006]. Benczes and Orzechowska-Waławska document how populist governments used economic policy, including family allowances and tax favors, to consolidate their coalitions [Benczes and Orzechowska-Waławska, 2024], and Ennser-Jedenastik shows that welfare chauvinism is a standard plank of populist radical-right platforms [Ennser-Jedenastik, 2018]. These works study welfare as an outcome of backsliding. This paper tests the opposite direction, whether welfare structure measured before erosion protects, with lead-lag tests that separate the two directions empirically.

The design gap is now precise. What the inequality-democracy literature has never done is to estimate the effect of a market-income inequality shock on democratic quality as a function of welfare institutions that preceded the shock, with a treatment that is not itself compressed by those institutions. Section 3 states that hypothesis; Section 4 builds the data; Section 5 fixes the estimation; Section 6 fixes the decision rules.

3 Theory and Hypotheses

3.1 The gate and its two paths

The gate claim is a claim about effect modification, and the entanglement problem reviewed in Section 2 forces the claim to be stated on the right treatment. Let m_{it} be market (pre-fisc) inequality in country i and year t , let d_{it} be disposable (post-fisc) inequality, and define the fiscal filter as the gap $g_{it} = m_{it} - d_{it}$. The gap is the share of market inequality that the fiscal system removes before the distribution reaches citizens. It is not a nuisance: it is the first mechanism of the gate, automatic stabilization, made visible. A market shock of size Δm produces a disposable shock of size $\Delta d = (1 - \phi(W))\Delta m$, where $\phi(W)$ is the filter, increasing in the scale and progressivity of the welfare state W .

The total effect of a market shock on liberal-democratic quality decomposes into two terms:

$$\frac{d\text{LibDem}}{dm} = \underbrace{\frac{\partial \text{LibDem}}{\partial d}(1 - \phi(W))}_{\text{fiscal filtering}} + \underbrace{\frac{\partial \text{LibDem}}{\partial m}\bigg|_{\bar{d}}}_{\text{behavioral transmission}},$$

where the first term is arithmetic: it operates even if no citizen changes any behavior, because less of the shock reaches the distribution politics responds to. The second term is the behavioral residue: the effect of the market shock on democratic quality holding the disposable path fixed, gated by constituency formation and credible pre-commitment. A research design that treats d as the treatment and W as the moderator conflates the two terms. A design that treats m as the treatment and measures g directly separates them. This paper registers the second design. Under the arithmetic-only account, the residual market effect is homogeneous in W once g is controlled, and the gate interaction disappears when the gap enters the specification. Under the behavioral account, the interaction survives gap control. Section 6 fixes which of these readings each pattern of results supports.

A simple sketch disciplines the functional form. Two periods. In period 0, the first democratic decade, society operates a universal service state of size w : services that a share $u(w)$ of households use, with $u' > 0$ and $u'' < 0$. In period 1, a market shock raises m , and a populist incumbent offers erosion: erosion delivers a private benefit B to the executive and captures the public services, worth λw to each user household. Household h supports erosion if its grievance $g_h(d) + v_h$ exceeds its stake, where $d = (1 - \phi(w))m$, grievance is increasing and convex in experienced inequality ($g' > 0$, $g'' > 0$), and v_h is the fear of elite capture in the sense of Acemoglu, Egorov, and Sonin, shifted up where inequality is high [Acemoglu et al., 2013]. Erosion proceeds if supporters cross a threshold. The marginal effect of the market shock on the erosion hazard is $\beta(w) = \gamma(1 - \phi(w)) - \delta u'(w) - \kappa A(w)$, where γ is the grievance channel, $\delta u'(w)$ is the constituency channel, and $\kappa A(w)$ is the pre-commitment channel, with $A(w)$ indexing program automaticity. With ϕ convex and u concave, $\beta'(w) < 0$ and $\beta''(w) > 0$: the gate exists, and protection is diminishing, so the marginal-effect curve is convex and threshold-like rather than linear. The sketch is a heuristic, not a structural model; its role is to discipline the functional forms reported in Section 6. We pre-commit to estimating the interaction linearly and with a quadratic term, and to plotting the inequality marginal effect at welfare-distribution percentiles so a threshold shape is visible if it is there.

3.2 Three mechanisms and their discriminating implications

Automatic stabilization. Universal public services sever household welfare from labor-market position. Public health care keeps medical costs off household budgets when earnings fall; public education keeps the next cohort's human capital accumulation independent of parental income shocks. This is decommodification applied to the post-1990 wave [Esping-Andersen, 1990], and its operational measure is the one the entanglement problem already produced: the market-minus-disposable Gini gap. The mechanism is direct, so its discriminating implication is direct. If the gate interaction is fully explained by the gap, that is, if including the gap and its interaction drives the residual inequality-by-welfare interaction to zero, stabilization is the entire mechanism. If the interaction survives gap control, a behavioral residue exists and must be attributed to the next two mechanisms.

Constituency formation. Programs create their own defenders [Pierson, 1993, Soss and Schram, 2007]. When the broad middle class uses public health and public education daily, it holds a material stake in those institutions: their quality, their budgets, their autonomy from the executive. Erosion threatens exactly that. An executive who captures the health ministry or the school system captures a service a household depends on, and a middle class that perceives

the capture as a loss of a valued public asset has a concrete, self-interested reason to resist. The microfoundation is the universalism-to-trust-to-legitimacy chain of Rothstein and Uslaner [Rothstein and Uslaner, 2005], applied at the margin of an inequality shock. The mechanism does not require civic virtue. It requires that the middle class has something at stake in the public sector. The discriminating implications: the gate should not exist where the welfare state is pension-heavy or means-tested, because the middle class is not enrolled; and the interaction should attenuate when spending composition, the share of public services in total social spending, and usage proxies are controlled.

Credible pre-commitment. When inequality is high, voters anticipate that elites will respond to redistributive threats by limiting redistribution, and the median voter’s demands go unmet; a populist who credibly promises redistribution can then win without being able to deliver it [Acemoglu et al., 2013]. Universal, automatic welfare programs pre-commit redistribution in advance: the institutional structure delivers services as a matter of routine, so the redistributive promise is not the exclusive property of an insurgent. The gate blocks the populist-promise equilibrium at its source, by removing the unmet demand that makes the promise decisive. The discriminating implication: the gate should be stronger where services are automatic, entitlement-based, indexed, and non-discretionary, and absent where generosity is discretionary and late-bought. The outcome-side literature has already documented the second half of this pattern, that incumbents under erosion expand targeted and chauvinist transfers while dismantling services [Szikra and Öktem, 2023, Lendvai-Bainton and Szelewa, 2021, Vanhuysse, 2006, Benczes and Orzechowska-Wa  awska, 2024, Ennsner-Jedenastik, 2018]; the design tests whether that late-bought generosity substitutes for the gate, which the hypothesis says it does not.

The three mechanisms share one implication that H5 states as a contrast: what gates is the universal service state, not welfare effort in general. Targeted cash transfers do not stabilize automatically, because they are discretionary; they do not enroll the middle class, because they are means-tested; and they do not pre-commit, because an incumbent can redirect them at will. Pension-heavy spending enrolls the elderly, not the young and the employed, and it is the component that distinguishing politicians can expand while dismantling services. Late-bought generosity is worse than neutral: it buys quiescence while institutions are dismantled.

3.3 The co-evolution of inequality, education, and democracy

The gate also explains a second set of facts: the co-evolution of inequality, education, and democratic quality in the post-1990 wave. Democratization has a robust effect on tax revenues and schooling but no robust average effect on inequality [Acemoglu et al., 2015]. The field fact, stated precisely in Section 2, is that the inequality-reduction leg of the democratic dividend does not materialize on average. The gate account explains why: fiscal expansion reduces inequality only where universal service institutions are strong enough to convert resources into distributional change and to keep leakage low. In thin-welfare democratizers, higher taxation without service delivery produces neither redistribution nor a middle-class constituency attached to public provision. The first-decade dividend, or its absence, then connects democratization to its own fate: countries whose first democratic decade delivered a visible, universal dividend entered the crisis years with a welfare buffer and a service constituency; countries where it did not entered them without either.

Education operates through the same institutional channel. The macro evidence that education causes democratization is weak [Acemoglu et al., 2005], and the gate account agrees at the transition margin. At the resilience margin the channel is different: education builds democratic resilience when it is absorbed into middle-class employment and enrolled in public services, because both give the educated a stake in the existing order. Where education expands without absorption, the

Campante-Chor mismatch operates, schooling combined with poor economic opportunities producing grievances and mobilization, transposed here from protest onset to graded erosion [Campante and Chor, 2012]. The redistributive politics of education [Ansell, 2010] and the cleavage structures documented by Gethin, Martinez-Toledano, and Piketty [Gethin et al., 2022] are the microfoundations of that channel. The hypothesis therefore predicts that education-without-absorption tightens the gate only where welfare is thin: the mismatch arm of the design.

3.4 Hypotheses

The theory yields five pre-registered hypotheses. Each is stated with its discriminating implication so that a confirming result can say which mechanism did the work, and a disconfirming result can say which account survives.

H1 (the gate). The within-country effect of a lagged market-inequality increase on the Liberal Democracy Index is negative and substantively large in thin-welfare countries and near zero or positive in strong-welfare countries; equivalently, the interaction of pre-fisc inequality with pre-determined welfare strength is positive and significant, under welfare-specific trend controls. The functional-form report pre-committed in Section 3.1 distinguishes a linear from a convex, threshold-like gate.

H2 (sequencing). Market-inequality surges and public-services retrenchment lead liberal-democracy declines by three to five years in thin-welfare countries, and no anticipatory service decline precedes erosion onset in high-welfare countries. The temporal order distinguishes the gate from the reverse-causal story in which erosion causes welfare decline [Szikra and Öktem, 2023].

H3 (the welfare-dependent dividend). Democratization reduces the top-10 percent income share, producing an inequality-reduction dividend over the first post-transition years, only in high-welfare-legacy countries; the size of that first-decade dividend predicts 2010–2022 resilience.

H4 (mechanisms). At least one discriminating mechanism result obtains: (a) the interaction attenuates toward zero once the market-minus-disposable gap and its interaction are controlled, supporting stabilization; (b) the interaction attenuates once spending composition and usage proxies are controlled, supporting constituency formation; (c) the interaction is stronger where services are automatic and absent for discretionary late-bought transfers, supporting pre-commitment; (d) the inequality-by-thin-welfare interaction raises V-Dem polarization, and education-without-absorption tightens the gate only where welfare is thin, consistent with the proximate-channel account in which universal institutions interrupt the polarization-to-erosion link that Rau and Stokes document [Rau and Stokes, 2025]. Arms (a) through (c) discriminate among the gate’s own mechanisms; arm (d) is a mechanism report placed in the proximate channel that the rival account already occupies, and its failure would not falsify the gate.

H5 (universalism contrast). The gate is produced by public services, health and education spending, not by total social spending, pension-heavy and transfer-dominated, and not by generosity acquired later through targeted or chauvinist transfers [Szikra and Öktem, 2023, Lendvai-Bainton and Szelewa, 2021, Vanhuysse, 2006, Benczes and Orzechowska-Wa  awska, 2024, Ennsner-Jedenastik, 2018]. Total social spending and late-bought transfers are placebo moderators: they do not gate.

The corollary states the asymmetry precisely. Welfare institutions built in the first democratic decade, before crises arrive, gate the inequality-erosion link. Welfare generosity purchased later by incumbents is not a substitute; it is part of the erosion playbook. Temporal priority of universal institution-building, not the level of welfare effort, is what protects. The cohort restriction to transitions beginning by 1998, the welfare window defined inside the first democratic decade, and the exclusion of cases whose erosion began within the first twelve years all operationalize the priority

claim, so that the pre-determined precedes the outcome it is used to explain.

4 Data and Measurement

The design is restricted to public OWID panels [Our World in Data, 2025]. OWID aggregates the underlying primary sources: V-Dem for regime and outcome measures [Lührmann et al., 2018, Coppedge et al., 2024], the World Bank’s Poverty and Inequality Platform for Gini coefficients, the World Inequality Database for top-income shares, WHO and World Bank EdStats for health and education spending, the ILO for labor-force indicators, and OECD for the placebo moderator. Every series is cited with its OWID grapher path in the data-availability table (Appendix A2), so any reader can reproduce the panel without proprietary data.

4.1 Sample

The treated population is post-1990 democratizers with transitions beginning between 1989 and 1998. Transition is dated from the OWID-hosted V-Dem Regimes of the World classification as the first year of electoral or liberal democracy sustained for at least two consecutive years, cross-checked against the OWID-hosted Boix-Miller-Rosato binary regime series [Boix et al., 2013, Lührmann et al., 2018]. If the cross-check must draw on the underlying BMR data rather than the OWID-hosted series, that is the paper’s single non-OWID input, disclosed here and in the registry. The cohort cap serves the pre-determination logic: for every retained case, the first democratic decade ends before the 2008 crisis and before the 2010–2022 resilience window that H3’s dividend must predict. The expected roster is 25 to 35 countries. The roster is fixed deterministically by the dating rule, printed in the registry and in Appendix A1, and no case is added or dropped after estimation begins. Countries with erosion onset within the first twelve years after transition are flagged; the specification with them excluded is pre-committed under Design 6(e) because welfare measured after erosion onset is post-treatment by construction. For comparison, and for the adjudication specification, we also construct the full sample of democracies over 1990–2022, which approximates and extends the Rau-Stokes population, whose sample restriction runs over 1995–2020, to the young democratizers that fell below the democracy threshold [Rau and Stokes, 2025]. The panel runs 1985–2023, leading the earliest transitions by four years to support event-study pre-trends.

4.2 Outcome

The V-Dem Liberal Democracy Index (`v2x_libdem`, 0–1), hosted on OWID, is the outcome in levels and in five-year differences [Our World in Data, 2025, Coppedge et al., 2024]. A continuous index is essential for graded erosion: the hypothesis concerns how much democratic quality is lost inside formally democratic regimes, and the post-1990 wave contains many episodes of substantial decline that stop short of breakdown. For the event-study designs, erosion onset is defined as the first year of a sustained three-year decline of at least 0.03 in the index, in line with the graded-decline approach of the measurement literature [Waldner and Lust, 2018]. Robustness to alternative onset codings is pre-committed.

4.3 The inequality treatment: pre-fisc by construction

The primary treatment is the WID pre-tax national-income top-10 percent income share, hosted on OWID. It is the theoretically preferred measure for the populism mechanism, since the fear of elite capture in Acemoglu, Egorov, and Sonin is a fear of the very rich [Acemoglu et al., 2013], and it

is the measure type, pre-tax WID shares, that Rau and Stokes use at the top end of their analysis [Rau and Stokes, 2025]. The secondary treatment is a market-income Gini, the WID pre-tax Gini or the PIP market-income Gini where available on OWID, with the source concept disclosed per country-year. The disposable-income Gini, in which the entanglement problem of Section 2 lives, is demoted to robustness: it appears in the specifications only to document the contrast and to construct the gap.

The gate’s first mechanism is measured directly. The market-minus-disposable Gini gap, git , is constructed wherever market and disposable series both exist, with concept tags recording the source combination (expected: WID pre-tax Gini minus PIP disposable Gini, and market-minus-disposable PIP where both exist). Interpolation behavior and per-series coverage flags are printed in the coverage matrix, because the gap is only as good as its two inputs and the design’s core identification move cannot rest on silently imputed observations. The gap is entered as a covariate and as a moderator in the mechanism and robustness designs.

The treatment is lagged by three years in the main specification, and distributed lags trace the one-to-six-year profile. The lag structure is fixed before estimation; it implements the three-to-five-year transmission window of H2 and separates the gate from the reverse-causal direction, in which erosion precedes the inequality surge.

4.4 The welfare measure and its windows

The gate variable W_i is the sum of domestic general government health expenditure (percent of GDP, WHO series) and government education expenditure (percent of GDP, World Bank EdStats), both hosted on OWID. It is an intentionally non-cash, services measure: public health and public education are the components the broad middle class uses, the components automatic in delivery, and the components an eroding executive must capture to take control of daily life. It excludes pensions and targeted cash by construction.

W_i is set to the average over the earliest available three-year window within years 0 to 8 after transition, and it is time-invariant thereafter. The actual calendar years of the window are documented per country in Appendix A1, because the WHO health series begins near 2000, so early transitions’ de facto measurement years are disclosed rather than assumed: a 1991 transition whose earliest available service data begin in 2000 is measured in years 9 to 11, flagged, and retained. The set of early transitions affected by the WHO start date is listed in the coverage matrix. The sensitivity pair is pre-committed: W measured at years $T+3$ to $T+7$ and at years $T+8$ to $T+12$ is reported alongside the primary earliest-window measure, so any window drift is visible rather than hidden. No country’s W is imputed.

The pre-determination logic is the heart of the identification, and it now has three legs. First, W is fixed before typical erosion onset: in the post-1990 cohort, erosion begins on average in the second decade after transition, and the cohort cap plus the early-erosion exclusion rule ensure the measurement window precedes onset in every retained case. Second, W precedes the 2008 crisis and the 2010–2022 resilience window for the entire cohort, so the dividend measure and the resilience outcome are temporally separated. Third, the welfare legacies are inherited rather than chosen: Eastern European democratizers walked into communist-era health and education systems of near-universal coverage, and Latin American and African democratizers into colonial-era public-service skeletons of varying depth. The variance in W is driven by the depth and maintenance of inherited systems through a chaotic first decade, not by long-run welfare design chosen under democratic deliberation.

4.5 Placebo moderators and mechanism variables

The universalism contrast of H5 needs a full-sample placebo. The OECD’s total social spending (percent of GDP) is pensions-heavy and transfer-dominated, and it is the natural contrast, but it exists only for OECD members, and most of the retained roster, including South Africa, Ukraine, Mongolia, Indonesia, and much of Latin America, would be excluded from an OECD-only placebo. The pre-committed primary placebo is therefore a full-sample transfer measure, the ILO World Social Protection Database or IMF Government Finance Statistics transfers, whichever OWID hosts for the roster; the OECD SOCX analysis is reported as a secondary subsample with its coverage limitation quantified. H5’s confirmation does not depend on the OECD-only subsample carrying the claim.

The mechanism variables implement Section 3.2. Stabilization is the gap. Constituency formation is probed with spending composition, the ratio of health-plus-education spending to total social spending, and with usage proxies, public secondary enrollment shares and the V-Dem public-services index, where OWID carries them. Pre-commitment is probed with an automaticity index constructed from entitlement, indexation, and non-discretionary funding features; its data availability is flagged in the coverage matrix, and if the index is unavailable for a substantial share of the roster, the pre-commitment arm is reported as a case-level supplement rather than a panel estimate. The proximate-channel arm uses the V-Dem political polarization index and the ILO educated-youth unemployment rate, the share of 15–24-year-olds with completed education who are unemployed, both hosted on OWID where available.

4.6 Controls

The core specifications include log GDP per capita (World Bank, via OWID) and an EU-accession indicator, with country fixed effects absorbing time-invariant regional differences and region-specific trends added in the robustness cells. Commodity dependence, measured as natural-resource rents as a share of GDP, enters the exclusion robustness check. No control is post-treatment with respect to the gate: W is pre-determined by construction, and the treatment is lagged. EU accession is entered both as a level control and, in the trend-controlled specification, as a level plus trend, because the accession process is the most salient welfare-building force in the Eastern European cases and the trend form is the form of the threat.

4.7 The divergent trajectories of the cohort

The phenomenon the design must explain is documented by the descriptive series that follow, with values from the development phase to be re-verified at every series URL in the coverage audit. Among the eleven cases with continuous data, liberal-democratic quality diverged sharply after 2010. Hungary fell from 0.754 in 2005 to 0.335 in 2022. Poland fell from 0.827 in 2010 to 0.417 in 2022. Estonia rose from 0.811 to 0.853 over 2010–2022, and Latvia, Lithuania, and Slovakia held their 2005 levels. Chile consolidated after 2000 and declined mildly after 2015; South Korea fell mid-decade and recovered, ending 2022 at 0.718. South Africa moved from 0.666 in 2010 to 0.610 in 2022, and Bulgaria and Romania fluctuated in an intermediate band. The same wave produced both the sharpest erosion and the most stable resilience in the same region.

Two descriptive facts about inequality discipline the theorizing. First, the eroded countries were not, on the disposable-income Gini, the most unequal in the group. Hungary’s Gini fluctuated between 0.27 and 0.35 over 2005–2016, with a trough of 0.270 in 2009; Poland’s fell from 0.38 in 2004 to 0.31 in 2016. Second, the resilient countries did not have exceptionally low inequality: Estonia’s Gini stayed between 0.31 and 0.35, Latvia’s between 0.34 and 0.39 (PIP, via OWID).

The cross-sectional correlation between inequality levels and erosion is weak in exactly the way Haggard and Kaufman describe [Haggard and Kaufman, 2021]. That fact does not contradict the materialist account, because the Rau-Stokes effect is about movements, not levels, and operates through within-country variation over time [Rau and Stokes, 2025]. It does mean the theory must be stated in shocks: the gate conditions the transmission of inequality increases, not the standing level of inequality. Section 5 is built for that statement.

The welfare numbers line up with the trajectories’ institutional reading. On the development-phase measure, the sum of health and education spending as a share of GDP runs from 10.3 in Lithuania, 9.6 in Hungary, 9.5 in Estonia, 9.0 in Slovakia, 8.9 in Poland, 8.4 in Latvia, 7.5 in Chile, 7.2 in South Africa, to 5.1 in South Korea; Bulgaria and Romania lack usable WHO health series in the early window on OWID (the coverage matrix reports the exact years). Hungary and Poland sit mid-pack on services and eroded at low inequality. If the gate claim is right in its strong form, their erosion must run through channels other than inequality, polarization and elite coalition formation first among them [Haggard and Kaufman, 2021, Svulik, 2019]. The welfare-gate hypothesis claims only that the inequality channel is gated by institutions, not that inequality is the only erosion channel. Whether mid-range service spending suffices to gate the inequality channel in Hungary and Poland, against the fiscal reality of pension-dominated welfare states in which the service component stagnated, is an open question that the interaction design, and only the interaction design, can answer. The marginal-effect reporting in Section 6 commits to reporting the inequality effect at welfare values that include the Hungarian and Polish cases, so the crucial cases are inside the support of the test.

4.8 Pre-determination and exogeneity probes

The identifying assumption deserves statement in its strongest form, and the probes that would break it are pre-committed. Post-1990 transitions are quasi-exogenous in timing: they cluster in 1989–1991 with the collapse of the Soviet bloc, in 1992–1994 with the Southern African and Central American settlements, and in the late 1990s with the remaining post-communist and Asian cases [Lührmann et al., 2018]. At the moment of transition, welfare legacies are inherited. The threat to the assumption is that the first democratic decade itself selects welfare depth: governments that consolidated well also maintained services. The design confronts this threat with five pre-committed probes.

First, balancing regressions: W is regressed on first-decade inequality changes and on fiscal-collapse indicators, inflation and output collapse, to show that W is not itself a product of the inequality dynamics the gate is meant to condition. Second, an instrumented version of Design 1: W is instrumented with legacy variables, communist-era schooling enrollment and colonial-era public-service coverage, with 2SLS estimates reported; the roster feasibility of each instrument is documented in the registry, and any shortfall is reported rather than papered over. Third, the within-Eastern-Europe estimate holds the communist legacy constant and varies only the depth of inherited health and education systems, where first-decade maintenance was driven by fiscal collapse and stabilization programs rather than democratic politics. Fourth, the placebo moderator splits the measure: if the protective variable were really democratic-quality selection, the pension-heavy total-spending measure would gate as well, and H5 predicts it does not. Fifth, the early-erosion exclusion rule removes the cases for which W is post-treatment by construction. Each probe addresses a specific version of the selection story, and Section 6 states what each failure would mean.

The coverage matrix closes the data section. Appendix A2 tabulates, for every roster country, every series, and every window: availability, first and last observed year, observation count, interpolation flag, and the OWID grapher URL. The effective estimation sample for each of the six designs

is printed from the matrix before estimation. If listwise deletion drops an effective sample below 25 countries, the minimum-detectable-effect analysis of Section 5.7 is recalibrated to the realized sample and reported, no imputation fills the gaps, and the confirmation criteria are interpreted at the realized power. The matrix is the answer to the coverage question a reviewer must ask of a 25-to-35-country design: every cell is either present with a URL or absent with a reason.

5 Research Design

All estimation is implemented in Python (linearmodels and statsmodels), with the staggered event studies estimated by Sun-Abraham via pyfixest and by Callaway-Sant’Anna via the R implementation through rpy2 or a documented manual implementation [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021]. Standard errors are clustered at the country level; the interaction tests use wild cluster bootstrap inference, given the modest number of clusters. The full code and the constructed panel are released with the registry deposit, so every number in the companion results paper is generated by the deposited code from the deposited data.

5.1 Design 1: the gate test

The core specification is a two-way fixed-effects regression with welfare-specific trends:

$$\text{LibDem}_{it} = \alpha_i + \delta_t + \beta_1 m_{i,t-3} + \beta_2 (m_{i,t-3} \times W_i) + \gamma_1 W_i t + \gamma_2 W_i t^2 + X'_{it} \theta + \varepsilon_{it},$$

where i indexes countries, t years, α_i and δ_t are country and year fixed effects, $m_{i,t-3}$ is the lagged pre-fisc inequality treatment, and W_i is the pre-determined welfare measure, absorbed by the country fixed effects in its own right with the interaction identified. The welfare-specific trends $W_i t$ and $W_i t^2$ are the identification change this registration makes relative to the designs reviewed in Section 2: any factor correlated with W and with democratic trajectories, EU accession, fiscal capacity, geopolitical alignment, is absorbed in the trend terms, and the gate is identified off deviations from welfare-specific trends. The gate hypothesis is $\beta_2 > 0$, large enough that the marginal effect $\beta_1 + \beta_2 W$ is near zero or positive in the high-welfare arm and strongly negative in the thin-welfare arm. We report the marginal effect at the 10th, 25th, 50th, 75th, and 90th percentiles of the welfare distribution, with wild cluster bootstrap confidence intervals, and the linear and quadratic functional forms pre-committed in Section 3.1. Because two-way fixed-effects estimators with differential timing can be biased under heterogeneous effects, we also estimate the interacted specification on five-year differences of the outcome and report Goodman-Bacon decompositions where timing variation is involved [Goodman-Bacon, 2021].

The critical control variant is the gap-controlled specification, which in the same equation adds $g_{i,t-3}$ and $g_{i,t-3} \times W_i$. Its reading is fixed in advance. If β_2 collapses toward zero once the gap and its interaction are included, the gate is fiscal filtering: the interaction was the first mechanism measured, and no behavioral residue exists. If β_2 survives, a behavioral residue exists and is attributed to constituency formation and pre-commitment, which Designs 5 and 6 then discriminate further.

5.2 Design 2: the adjudication specification

To make the reconciliation claim directly, we estimate the baseline inequality main effect without any interaction, on the full democracy sample and on the young-democracy sample, and then introduce the interaction. The protocol commits to reading the two specifications jointly. A weak or insignificant main effect with a strong and significant interaction is the pattern that reconciles Rau and Stokes with Haggard and Kaufman [Rau and Stokes, 2025, Haggard and Kaufman, 2021],

because it says the average effect hides institutional heterogeneity rather than being absent. A strong unconditional main effect that survives the interaction is instead evidence for the unconditional materialist account. A null everywhere is evidence for the purely coalitional account. The three-way reading is pre-committed before estimation, so the adjudication cannot be renegotiated after the results are in.

5.3 Design 3: erosion-onset event studies and sequencing

We stack erosion onsets among post-1990 democratizers and estimate dynamic coefficients relative to onset, separately for the high-welfare and thin-welfare arms defined by a median split of W . The estimators are Callaway-Sant’Anna and Sun-Abraham, chosen because onsets are staggered across the cohort and treatment effects are plausibly heterogeneous [Callaway and Sant’Anna, 2021, Sun and Abraham, 2021]; the Goodman-Bacon decomposition documents the composition of any two-way fixed-effects benchmark [Goodman-Bacon, 2021]. Pre-trend tests are pre-committed and reported for both arms. The event-study coefficients trace the Liberal Democracy Index, market inequality, and public-services spending in the eight years before and after onset. The confirmation pattern for H2: in the thin-welfare arm, inequality and service retrenchment rise before the index declines; in the high-welfare arm, no such pre-onset movement appears. The lead-lag structure is the primary defense against reverse causality. If erosion caused welfare decline, services would fall only after onset; the hypothesis requires service decline to lead erosion in the thin-welfare arm and to be absent in the high-welfare arm. This also separates the gate’s direction from the outcome-side literature that documents welfare transformed under backsliding [Szikra and Öktem, 2023, Lendvai-Bainton and Szelewa, 2021, Vanhuyse, 2006, Benczes and Orzechowska-Wałaszewska, 2024].

5.4 Design 4: the democratization event study and the dividend

We align countries at the transition year T and estimate the dynamic path of the top-10 percent income share, public services, and education over the two decades after transition, separately by pre-determined welfare legacy, above or below the sample median. The dividend window is fixed at years 0 to 7 after transition, so the dividend precedes the 2008 crisis and the 2010–2022 resilience window for every retained case. The confirmation pattern for H3: the top-10 percent share declines in years 0 to 7 only in high-welfare-legacy countries, and the size of that first-decade decline predicts the 2010–2022 change in the Liberal Democracy Index, in a country-level regression that reports the dividend-resilience slope with its confidence interval. This extends the democratic-dividend result [Acemoglu et al., 2015] by conditioning its inequality leg on welfare institutions and by connecting the dividend to subsequent resilience, with the dividend measured before the resilience outcome.

5.5 Design 5: mechanism arms

(a) Stabilization: the gap-controlled variant of Design 1, whose reading is fixed in Section 5.1. (b) Constituency formation: the interaction is re-estimated with spending composition, the health-plus-education share of total social spending, and usage proxies entered as additional moderators and controls; attenuation is the pattern that assigns work to the constituency channel. (c) Pre-commitment: the automaticity index is interacted with the treatment in the same design; the gate should be stronger where services are automatic. (d) Proximate channels: the inequality-by-thin-welfare interaction is estimated on the V-Dem polarization index, and a triple interaction of inequality, thin welfare, and educated-youth unemployment tests whether the gate tightens where education expands without absorption [Campante and Chor, 2012]. Arms (a) through (c) are the mechanism discrimination the theory requires; arm (d) is a mechanism report in the proximate

channel already occupied by the rival account, and it is interpreted with that status, as Section 3.4 pre-commits.

5.6 Design 6: robustness

Seven cells are pre-committed, and every one is reported, including the ones that fail. (a) Alternative treatments: the market-income Gini for the top-10 percent share, and the disposable Gini entered only to document the contrast that motivates the design. (b) Alternative welfare measures: health and education entered separately; the sum entered continuously and as an above/below-median split. (c) Population weights. (d) Exclusion of the commodity-boom countries, identified by natural-resource rents above a threshold. (e) The within-Eastern-Europe estimate, holding the communist legacy constant; the early-erosion exclusion, dropping or recoding cases with erosion onset within the first twelve years; and the EU-accession trend control. (f) Wild cluster bootstrap confidence intervals for the interaction and the marginal effects. (g) Placebo moderators: total social spending from the full-sample transfer measure, the ILO World Social Protection Database or IMF Government Finance Statistics series, and the OECD SOCX measure on its OECD subsample; plus a placebo interaction of W with a within-country variable unrelated to inequality, population age structure, which the gate account predicts is null. The placebo cells are the falsification of the H5 contrast: a gate reproduced by pensions or by targeted or chauvinist transfers disconfirms the universalism claim.

5.7 Power, minimum detectable effects, and the registry

Power for an interaction test in a panel of 25 to 35 countries with serially correlated outcomes is the binding constraint, and the criteria in Section 6 are calibrated to it. Before estimation, a simulation-based minimum-detectable-effect analysis is run on the actual panel structure: a two-way fixed-effects DGP with the realized country roster, the realized W distribution, and the realized treatment moments, under serial correlation of the outcome at ρ 0.7, 0.8, and 0.9 and effective N of 25, 30, and 35. The analysis reports the minimum detectable β_2 and the minimum detectable marginal effect at the 10th percentile of W , at 80 percent power, and the report is deposited with the registry. Two consequences follow and are accepted in advance. The study is powered to detect a large conditional effect, which is what the theory predicts: the Section 3 sketch implies near-complete transmission where welfare is thin and near-zero transmission where it is thick. And the confirmation criteria are stated for effects of that size; a weak gate would go undetected, and the paper will say so rather than claim a null.

The registry closes the design. The protocol, the roster, the coverage matrix, the MDES analysis, and the deposited code are timestamped at OSF before any estimation. The primary specification is Design 1 with the pre-fisc treatment, the earliest-window W , and welfare-specific trends. The decision rule, fixed in advance: the gate holds if the interaction is significant at $p < 0.05$ and the marginal-effect pattern is confirmed in the primary specification and in at least five of the seven robustness cells, with failures reported and interpreted. No specification is added after estimation; no specification is dropped; the deposited code generates every number in the companion paper.

6 Expected Results and Falsification Criteria

The protocol commits ex ante to the following inference rules, so that the test can adjudicate between the accounts.

Confirmation of the welfare-gate hypothesis requires five conditions. (a) A significant positive interaction between lagged pre-fisc inequality and pre-determined welfare strength on liberal-democracy trajectories, under welfare-specific trend controls, with the within-country inequality effect near zero or positive in the high-welfare arm and strongly negative in the thin-welfare arm: the marginal effect at the 10th and 25th percentiles of W is negative with a confidence interval excluding zero, and the marginal effect at the 90th percentile is not significantly negative. The gate holds if this pattern appears in the primary specification and in at least five of the seven robustness cells, with failures reported and interpreted. (b) Sequencing: market-inequality surges and public-service retrenchment lead liberal-democracy declines by three to five years in thin-welfare countries, no anticipatory service decline precedes erosion onset in the high-welfare arm, and pre-trend tests pass in both arms. (c) The welfare-dependent dividend: the top-10 percent income share declines in years 0 to 7 after democratization only in high-welfare-legacy countries, and the size of that first-decade decline predicts the 2010–2022 change in the Liberal Democracy Index. (d) At least one discriminating mechanism result from Design 5, with the readings fixed in Section 5.1 and stated again here: the interaction collapses when the gap and its interaction are controlled, and stabilization is the mechanism; or the interaction attenuates with composition and usage controls, and constituency formation is the mechanism; or the interaction is concentrated in automatic programs and absent for discretionary late-bought transfers, and pre-commitment is the mechanism. (e) The universalism contrast: public services gate; the full-sample transfer placebo and the OECD total social spending measure do not, in their respective samples.

Disconfirmation is equally sharp, and each outcome has a clear interpretation in the literature. A homogeneous inequality effect across welfare arms, or an insignificant or wrong-signed interaction in the primary specification, falsifies the gate and favors the unconditional materialist account [Rau and Stokes, 2025, Houle, 2009]. An interaction that collapses once welfare-specific trends are added falsifies the gate in favor of a selection reading in which a third factor drives both welfare maintenance and democratic resilience. Reversed sequencing, with service decline following erosion onset, favors the outcome-side transformation literature [Szikra and Öktem, 2023, Lendvai-Bainton and Szelewa, 2021, Vanhuysse, 2006, Benczes and Orzechowska-Wacławska, 2024] over the gate. A gate reproduced by the pension or targeted-cash placebo disconfirms the universalism claim, H5, even if H1 holds. A dividend that appears in thin-welfare countries disconfirms the welfare-dependent-dividend claim, H3. Every one of these outcomes adjudicates among the accounts, so the study is a genuine three-way test and a null result is itself a contribution: it would mean the unconditional materialist or the purely coalitional account survives the institutional test.

Two interpretive commitments deserve emphasis because they are the ones a design can quietly reverse. First, the arithmetic-only reading is pre-committed. Under the arithmetic-only account, in which automatic stabilization filters inequality but institutions change no behavior, the gate interaction in the gap-controlled specification disappears completely, and the residual market effect is homogeneous in W . Under the behavioral account, a residual interaction survives. The results are read through that dichotomy, and the two mechanisms that survive the gap control, constituency formation and pre-commitment, are then assigned by the Design 5 attenuation patterns. Second, the marginal-effect reporting standard: the gate claim lives or dies on the inequality marginal effect at the welfare-distribution percentiles, not on the interaction coefficient alone, because a significant interaction with a negative marginal effect everywhere would not constitute a gate. The percentile plot and the Hungarian and Polish values inside the support are part of the report, not an afterthought.

7 Discussion

The welfare-gate hypothesis is offered as a resolution to a dispute that has been stuck because its two protagonists estimate different objects, and one of them measures its treatment through the very institutions the dispute is about. The value of the design lies in what each possible outcome would teach. We discuss three central readings before the data arrive.

Reconciling the two accounts. If H1 and H2 hold, the Rau-Stokes inequality effect [Rau and Stokes, 2025] and the Haggard-Kaufman economic skepticism [Haggard and Kaufman, 2021] are not in tension. The inequality effect in the global sample is carried by the thin-welfare democracies: the post-communist laggards, the Latin American service skeletons, the Southern African cases where public services reach only part of the population. In the high-welfare arm, inequality rises without erosion, which is why close comparison of resilient cases finds no economic story. The case-based account studied mostly the backsliders, Hungary, Poland, Turkey, and the statistical account studied mostly the pooled average. Both were right about their effective samples and wrong about the population. The gate also disciplines the level-versus-shock distinction that the descriptive data force on us: Hungary and Poland eroded at low inequality, so the inequality channel cannot be their whole story; the gate claim is that where inequality shocks did occur, their transmission depended on institutions. That is a narrower claim than the materialist account makes, and the design is built to test exactly that claim.

The pre-fisc correction changes what a confirming result would mean for the materialist account itself. If the gate survives on the pre-tax top-10 percent share and on the market Gini, the inequality-erosion link is real but institutionally conditioned, which is the reconciliation. If the gate appears on the disposable Gini and dissolves on the pre-fisc measures, the Rau-Stokes main effect is partly fiscal filtering after all: the disposable Gini fell in high-welfare countries partly because their fiscal systems compressed it, and the “protective” welfare state and the “compressed” treatment are the same object measured twice. The design’s contrast specifications are built so the results can say which reading holds.

The democratic dividend, conditioned. The most consequential extension is to the democratic-dividend literature. Acemoglu, Naidu, Restrepo, and Robinson established that democratization raises taxation and schooling, and found no robust average inequality reduction, with imprecise estimates and heterogeneity they could not fully explain [Acemoglu et al., 2015]. The welfare-gate account supplies a candidate conditioner: the inequality-reduction leg of the dividend materializes only where universal service institutions are strong enough to convert fiscal expansion into distributional change. In thin-welfare democratizers, higher taxation without service delivery produces neither redistribution nor a middle-class constituency attached to public provision. The first-decade dividend, or its absence, is then the mechanism that connects democratization to its own fate: countries whose first democratic decade did not deliver the visible, universal dividend entered the crisis years with no welfare buffer and no service constituency, precisely the configuration that H1 predicts erodes under inequality pressure. If H3 and H4 hold, the dividend literature and the erosion literature are one literature: the dividend is the resilience mechanism, and its failure is the erosion mechanism. This also reframes the education-and-democracy null [Acemoglu et al., 2005]: education produces democratic resilience through absorption into middle-class employment and enrollment in public services, not through values socialization alone. Where education expands without absorption, the Campante-Chor mismatch [Campante and Chor, 2012] operates, and the educated young become an erosion resource rather than a democratic one. The cleavage-structure results of Gethin, Martinez-Toledano, and Piketty [Gethin et al., 2022] and the redistributive politics of education in Ansell [Ansell, 2010] are the microfoundations of that channel. We repeat the caution of Section 6: all of this is conditional on the design confirming, and the design is

registered precisely so that a reader can tell the difference between a confirming and a disconfirming pattern without trusting the authors' later description of what they intended.

The universalism contrast and the policy reading. The sharpest claim in the paper is H5: total social spending and late-bought generosity do not gate. If H5 holds, the policy implication is not spend more on welfare but build universal service institutions early. For the post-1990 cohort and for any future democratization wave, the testable prescription is that the first democratic decade is the window in which universal health and education systems must be constructed, because legitimacy and constituency are built by the institutions that exist when the first crisis arrives. Welfare effort purchased later, targeted cash, chauvinist family allowances, functions as pacification [Vanhuyse, 2006] and as an instrument of the eroding incumbent [Szikra and Öktem, 2023, Lendvai-Bainton and Szelewa, 2021, Benczes and Orzechowska-Waławska, 2024]. The distinction between universal services and total social spending is the distinction between institutions that create their own defenders [Pierson, 1993, Soss and Schram, 2007, Rothstein and Uslaner, 2005, Korpi and Palme, 1998] and expenditures that buy quiescence.

South Africa is the sharpest challenge to the gate's thin-welfare arm. It combines the cohort's highest inequality, a Gini near 0.65 at its 2005 peak and still 0.61 in 2010, with only modest liberal-democratic decline, from 0.666 in 2010 to 0.610 in 2022. A purely materialist account would predict more erosion; the gate account reads the pattern as delayed or partial transmission, with the state-capture scandals of the 2010s as the first installment. The timing tests in Design 3 are what distinguish these readings, and the protocol commits to reporting South Africa's residual explicitly.

8 Limitations

Five limitations are intrinsic to the design, and we state them as such rather than as future work.

Small N and statistical power. The treated sample is 25 to 35 countries, and the interaction is a difference-in-differences of a difference-in-differences: the effective variation is the within-country association between inequality and liberal democracy, compared across welfare arms. Power for interaction tests is low in samples of this size, and the clustered standard errors rest on a modest number of clusters. The pre-committed MDES analysis defines what the design can detect before estimation, and the confirmation criteria are stated for the large conditional effects the theory predicts. The study will not detect a weak gate, and the results paper will say so if the confidence intervals cannot exclude one.

The welfare measure is a proxy. Public health and education spending as a share of GDP measures fiscal effort, not universality, not quality, and not usage by the middle class. Two countries with identical spending shares can differ in whether public hospitals serve the well-off or only the poor, and the mechanism in H1 is about usage. The proxy bias is conservative: measurement error in the gate variable attenuates the interaction and biases against confirmation. Composition and usage controls partially discipline the proxy, but the ideal test would require usage data that OWID panels do not fully carry, and the coverage matrix will say exactly how far the usage proxies reach.

Identification of the gate. The welfare measure is pre-determined relative to erosion onset, but it is not randomly assigned. The first democratic decade is itself a period of democratic politics, and governments that consolidated well also maintained services. The balancing regressions, the legacy-instrument 2SLS, the within-Eastern-Europe estimate, the placebo moderator, and the early-erosion exclusion each address a specific version of this threat, and the welfare-specific trends absorb the shared-trend version, but none eliminates the possibility that a third factor, state capacity, geography, or geopolitical alignment, drives both welfare maintenance and democratic resilience. EU accession is the most salient such factor for the Eastern European cases, and it was itself

a welfare-building force in the first decade, which means the gate and the accession channel are partially entangled by construction. The time-invariant nature of the moderator means the cross-sectional weight in the interaction is correlational by design; the identification claims are about the within-country inequality dynamics that the trends and lags discipline.

Measurement of the outcome and the treatment. The Liberal Democracy Index is expert-coded, and its inter-temporal comparability is contested [Waldner and Lust, 2018]. The Gini coefficient’s cross-country comparability is limited by differing underlying surveys and income concepts; PIP and WID harmonization, while the best available, does not remove the issue. Both biases are plausibly classical in this design, attenuating main effects and interactions alike, but the threat to the sequencing tests is asymmetric: if erosion onset is misdated, the lead-lag structure in Design 3 shifts, and the confirmation pattern for H2 could be manufactured by measurement timing. We address this with the three-year sustained-decline coding rule, robustness to alternative onset codings, and the requirement that the sequencing pattern differ across welfare arms, which a pure measurement artifact would not produce.

The placebo subsample problem. The sharpest contrast, H5, requires a full-sample placebo, and the full-sample transfer measure is the pre-committed answer. The OECD SOCX measure exists only for OECD members, and an H5 test run only on that subsample could not carry the paper’s strongest claim; the OECD-only analysis is reported as a secondary subsample with its coverage limitation quantified. If the ILO or IMF full-sample series proves unavailable on OWID for a substantial share of the roster, that finding is itself a registered limitation: the universalism contrast would then rely on the composition contrast within the health-plus-education measure, health versus education entered separately, and on the within-country contrast between discretionary and automatic spending, and the paper would say so.

9 Conclusion

The post-1990 democratizers are the population on which the dispute about democratic erosion should be adjudicated, and they are the population the global statistical literature has excluded. This paper has argued that the adjudicating variable is the welfare state, specifically the universal service state built in the first democratic decade, and that the field’s flagship treatment, disposable-income inequality, is compressed by the very institutions the gate hypothesis implicates. The hypothesis is precise: market-inequality surges erode young democracies only where welfare institutions are too thin to absorb them, the fiscal filter is measured directly as the market-minus-disposable gap, and the inequality-reduction leg of the democratic dividend materializes only behind the gate. The design is fixed: an OWID-only panel, a pre-determined welfare measure with documented windows, a pre-fisc treatment, welfare-specific trend controls, staggered event studies with estimators built for differential timing, exogeneity probes, a fixed roster with a coverage matrix, a power analysis, and a timestamped registry. The descriptive data show why the adjudication matters: the same wave produced both the sharpest erosion and the most stable resilience, and the eroded countries were not the most unequal. If the evidence confirms the gate, the reconciliation of the materialist and coalitional accounts is institutional: inequality matters, but its effect is conditional on institutions that half the post-1990 wave never built, and part of the materialist account’s pooled effect is the fiscal filtering of its own treatment. If it disconfirms, the unconditional accounts stand, and the adjudication is still progress. The results paper will report against the criteria fixed here, and the registry entry will make the difference between the two visible to any reader.

Appendix

A1. Candidate roster and transition years. The roster is fixed deterministically from the OWID-hosted V-Dem Regimes of the World series, cross-checked against Boix-Miller-Rosato, and is printed in full, country by country with transition year and dating source, in the registry and in the supplementary materials of the companion results paper. No case is added or dropped after estimation begins. An illustrative subset of eleven of those cases, Bulgaria, Chile, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Slovakia, South Africa, and South Korea, is listed here; the full roster is produced by the dating rule of Section 4.1, and its coverage appears in Appendix A2.

A2. Data-availability and coverage matrix. Country by variable by window: availability, first and last observed year, observation count, interpolation flag, OWID grapher URL.

A3. Minimum-detectable-effect analysis. Simulation results on the realized panel under ρ 0.7, 0.8, 0.9 and effective N 25, 30, 35.

A4. Registry deposit checklist. Protocol text, roster, coverage matrix, MDES report, code, panel construction script, decision rules, dated OSF deposit.

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